

UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI
DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY



BACHELOR THESIS

By

NGUYEN DUY NGOC

Title:

**Development of an Internal AI Chatbot
for Company Knowledge Retrieval**

External Supervisor: Dr. Trần Đình Chí

Internal Supervisor: Dr. Trần Hoàng Tùng

Hanoi, July 2026

To whom it may concern,

I, **Trần Đình Chí** (supervisor's name), certify that the thesis/ internship report of Mr/Ms. **Nguyen Duy Ngoc** (student's name) is qualified to be presented in the Internship Jury 2025–2026.

Hanoi, __/__/____

Supervisor's signature

Contents

List of Abbreviations	iv
List of Tables	v
List of Figures	vi
Acknowledgements	vii
Abstract	viii
1 Introduction	1
1.1 Context and Motivation	1
1.2 Problem Statement	2
1.3 Literature Review	2
1.4 Contribution of This Work	5
2 Objectives	7
3 Materials and Methods	8
3.1 System Architecture	8
3.2 Knowledge Base Construction	9
3.2.1 Document Authoring	9
3.2.2 AI-Generated Documents	10
3.2.3 Comparison with Related Datasets	10
3.3 Document Processing and Indexing	11
3.3.1 Data Understanding	11
3.3.2 Chunking Strategy	12
3.3.3 Embedding	13
3.3.4 Vector Store	14
3.4 Hybrid Retrieval: BM25 + Vector Search with RRF	14
3.4.1 Motivation	14
3.4.2 BM25 Retrieval	14

3.4.3	Reciprocal Rank Fusion	14
3.5	Answer Generation	15
3.5.1	Model	15
3.5.2	Prompt Format	15
3.6	QLoRA Fine-Tuning Experiment	16
3.6.1	Training Configuration	16
4	Results and Discussion	18
4.1	Evaluation Metrics	18
4.1.1	Retrieval Metrics	18
4.1.2	Answer Generation Metrics	18
4.2	Retrieval Comparison: Pure Vector vs. Hybrid BM25+Vector	19
4.3	End-to-End System Demonstration	21
4.4	QLoRA Fine-Tuning: Negative Result Analysis	23
4.4.1	Training Data Quality	23
4.4.2	Evaluation Results	23
4.4.3	Root Cause and Lessons Learned	23
4.5	Discussion	24
4.5.1	Effectiveness of Hybrid Retrieval	24
4.5.2	RAG vs. Fine-Tuning: Practical Implications	24
4.5.3	Limitations	25
4.5.4	Comparison with Related Systems	26
5	Conclusion and Perspective	27
5.1	Achievements	27
5.2	Key Findings	27
5.3	Perspectives and Future Work	28
A	Key Code Listings	31
A.1	Hybrid Retriever (src/hybrid_retriever.py)	31
A.2	Prompt Template (src/rag_pipeline.py)	32
B	Evaluation Results (Full)	33

List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BM25	Best Match 25
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
GGUF	GPT-Generated Unified Format
GPU	Graphics Processing Unit
HR	Human Resources
KB	Knowledge Base
LLM	Large Language Model
LoRA	Low-Rank Adaptation
NLP	Natural Language Processing
QLoRA	Quantized Low-Rank Adaptation
RAG	Retrieval-Augmented Generation
RRF	Reciprocal Rank Fusion
SLM	Small Language Model
UI	User Interface
USTH	University of Science and Technology of Hanoi
VRAM	Video Random Access Memory

List of Tables

3.1	Manually authored Vietnamese HR policy documents (8 files).	9
3.2	Comparison of this work’s knowledge base with related QA datasets.	11
3.3	Chunk statistics for the 20-document Vietnamese HR knowledge base.	13
3.4	QLoRA fine-tuning experimental setup.	17
4.1	Top-1 retrieved document: pure vector vs. hybrid retrieval on five Vietnamese HR queries.	19
4.2	Retrieval metrics on 10 Vietnamese HR queries: pure vector baseline vs. hybrid BM25+vector with RRF.	20
4.3	End-to-end chatbot demonstration: 5 Vietnamese HR queries with timing. . . .	21
4.4	Content analysis of QLoRA training dataset (3,343 rows).	23
B.1	Full retrieval evaluation on 10 Vietnamese HR queries.	33

List of Figures

- 3.1 End-to-end system architecture. User query flows through embedding, hybrid BM25 + vector retrieval, Phi-3 chat prompt construction, and GGUF inference to produce a grounded Vietnamese answer. 8
- 3.2 Distribution of 20 Vietnamese HR documents by topic category. 8 documents were manually authored; 12 were generated via Minimax-M2.7 API. 10
- 3.3 Hybrid retrieval pipeline. BM25 and vector search independently rank all corpus documents; scores are fused via RRF and the top-5 results are selected as retrieval context. 15
- 4.1 Top-1 retrieval accuracy on 10 Vietnamese HR queries: pure vector baseline (60%) vs. hybrid BM25+vector with RRF (100%). 21
- 4.2 CLI chatbot session: user queries Vietnamese HR policy and receives a grounded answer with source attribution. 22

Acknowledgements

I would like to express my sincere gratitude to all those who supported me throughout the completion of this thesis.

First and foremost, I am deeply grateful to my External Supervisor, Dr. Trần Đình Chí, for his insightful guidance, constructive feedback, and continuous encouragement during the research process.

I also sincerely thank my Internal Supervisor, Dr. Trần Hoàng Tùng, for his academic support and for providing the institutional framework that made this project possible within the University of Science and Technology of Hanoi.

I extend my appreciation to the faculty and staff of the Department of Information and Communication Technology at USTH for the knowledge and resources they provided throughout my studies.

Finally, I thank my family and colleagues for their patience and moral support during the writing of this thesis.

Abstract

This thesis describes a fully local, offline Retrieval-Augmented Generation (RAG) chatbot that answers Vietnamese Human Resources (HR) policy questions. Employees can query internal company policies in Vietnamese without sending data to any cloud service, preserving privacy and avoiding per-query API costs.

The knowledge base contains 20 Vietnamese HR policy documents covering leave policy, employment contracts, salary, social insurance, and other workplace regulations. Eight documents were written by hand and twelve were generated using the Minimax-M2.7 language model API. Documents are split into 600-character overlapping chunks using a recursive character text splitter, embedded with `paraphrase-multilingual-MiniLM-L12-v2` (384-dimensional vectors), and stored in a persistent ChromaDB vector database.

The main technical contribution is a hybrid retrieval pipeline combining BM25 keyword search with vector cosine similarity, merged via Reciprocal Rank Fusion (RRF). On a 10-query evaluation set, hybrid retrieval achieves 100% top-1 accuracy and MRR of 1.00, compared to 60% accuracy and MRR of 0.72 for pure vector search. The improvement is most pronounced on short (3–5 token) Vietnamese queries, where keyword overlap is more discriminative than embedding similarity. Answer generation uses Phi-3-Mini-4k-instruct in GGUF Q4_K_M format, running locally via `llama-cpp-python`.

A separate QLoRA fine-tuning experiment on 3,343 synthetic question-answer pairs produced catastrophic hallucinations due to training data contamination (only 37% domain-relevant content). This negative result is reported with root cause analysis, highlighting the critical role of data curation in domain fine-tuning.

Keywords: Retrieval-Augmented Generation, Large Language Model, Vietnamese NLP, BM25, vector search, knowledge retrieval.

1. Introduction

1.1 Context and Motivation

Large Language Models (LLMs) have opened practical possibilities for automating knowledge retrieval in enterprise environments [1]. Companies accumulate large volumes of internal documentation — employee handbooks, policy manuals, standard operating procedures — and employees spend considerable time searching through these documents for specific answers.

In Vietnamese companies, HR policy documents are typically long (50–200 pages), verbose, and available only in PDF or printed format, requiring employees to spend 30–60 minutes searching for a specific policy answer. A 2023 survey by VietnamWorks found that 67% of Vietnamese office workers reported spending more than 30 minutes per week searching for internal company policies. This productivity loss is compounded by the fact that HR policies are frequently updated, and employees may unknowingly reference outdated versions.

Cloud-based solutions such as ChatGPT or Gemini could theoretically address this problem, but introduce serious concerns for enterprise deployment:

- **Data privacy:** Sensitive HR data — including salary structures, disciplinary procedures, and employee personal information — must be transmitted to third-party servers. For Vietnamese companies subject to the 2023 Personal Data Protection Decree (Decree 13/2023/ND-CP), this may constitute a regulatory violation.
- **Cost scalability:** Per-query API costs (approximately \$0.002–\$0.06 per query for GPT-4-class models) scale linearly with company-wide usage. A company with 500 employees making 10 queries per day would incur \$3,000–\$90,000 annually in API costs alone.
- **Grounding:** Cloud LLMs generate answers from parametric knowledge trained on public data. They cannot reliably cite specific clauses from the company’s internal documents, and may produce plausible but incorrect answers (hallucinations) about policies they have never seen.

Retrieval-Augmented Generation (RAG) [1] offers a principled solution to all three concerns: rather than relying solely on the LLM’s parametric knowledge, the system first retrieves relevant document passages from a local knowledge base and injects them into the prompt context. This grounds the generated answer in verifiable source material, enables the use of a smaller locally-

run model (since the retrieval step compensates for limited parametric knowledge), and keeps all data on-premises.

1.2 Problem Statement

While the RAG architecture addresses the core privacy and grounding concerns, building a working Vietnamese HR chatbot involves several practical challenges that this work targets:

1. **Information access latency:** Vietnamese employees cannot quickly access HR policy answers without manually reading through long documents. The absence of a structured query interface forces employees to rely on word-of-mouth or direct HR department inquiries, creating bottlenecks.
2. **Privacy and cost barriers:** Cloud LLMs are unsuitable for internal HR queries due to Vietnamese data protection regulations, organizational security policies, and per-query API costs that do not scale economically for company-wide deployment.
3. **Retrieval failure on short queries:** Standard vector cosine similarity search performs poorly on short Vietnamese queries (3–6 tokens), where keyword overlap is more discriminative than embedding similarity. This is a known weakness of dense retrieval for short queries in morphologically rich languages.
4. **Training data scarcity:** Fine-tuning a domain-specific model requires curated, high-quality training data. Vietnamese HR content is scarce online, and automated web crawling produces contaminated datasets mixing Vietnamese labor law with unrelated domains (Japanese employment regulations, US healthcare policy).
5. **Hardware constraints:** Many Vietnamese SMEs do not have access to high-end GPU servers. The system must run on consumer-grade hardware (8 GB RAM, integrated GPU or entry-level discrete GPU) to be practically deployable.

1.3 Literature Review

Retrieval-Augmented Generation. Lewis et al. [1] introduced RAG as a general architecture combining a dense retrieval component with a sequence-to-sequence generator. The retriever uses a pre-trained bi-encoder to fetch relevant passages from an external knowledge corpus, and the generator (originally BART) conditions on both the question and the retrieved passages to produce grounded answers. Two variants were proposed: *RAG-Sequence*, which uses the

same set of retrieved documents for the entire output sequence, and *RAG-Token*, which can retrieve different documents for each output token. This architecture has become the dominant approach for knowledge-intensive NLP tasks, as it decouples the knowledge source from the model parameters — allowing the knowledge base to be updated without retraining the model.

Subsequent work has refined the RAG paradigm. Gao et al. (2023) proposed *Self-RAG*, where the model decides when to retrieve and evaluates its own outputs for factual accuracy. Asai et al. (2024) introduced *REPLUG*, which treats the retriever as a plug-in module that can be fine-tuned independently. For enterprise applications, the key advantage of RAG over pure fine-tuning is auditability: every generated answer can be traced to specific source passages, enabling compliance verification.

Lexical Retrieval: BM25. BM25 (Best Match 25) [2] is a probabilistic ranking function from the family of bag-of-words retrieval models. It scores documents based on term frequency (TF) and inverse document frequency (IDF), with two key innovations over TF-IDF: (1) a saturation function that prevents high term frequencies from dominating the score, controlled by parameter k_1 (typically 1.2–2.0); and (2) document length normalization controlled by parameter b (typically 0.75), which penalizes very long documents that would otherwise accumulate high scores simply due to their length.

The BM25 scoring function for a query Q containing terms $q_1, \dots, q_n$ against a document D is:

$$\text{BM25}(D, Q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot (1 - b + b \cdot \frac{|D|}{\text{avgdl}})} \quad (1.1)$$

where $f(q_i, D)$ is the frequency of term q_i in document D , $|D|$ is the document length, and avgdl is the average document length in the corpus. Despite its simplicity and age (first proposed in the 1990s), BM25 remains highly competitive for short, keyword-focused queries and serves as a strong baseline in modern information retrieval benchmarks, including the MS MARCO and BEIR benchmark suites.

Hybrid Retrieval and Reciprocal Rank Fusion. A well-documented limitation of dense retrieval is its poor performance on queries where exact lexical matching is critical — for instance, when the query contains a specific policy number, a proper noun, or a short keyword phrase. Conversely, BM25 fails when the query uses synonyms or paraphrases that differ lexically from the document vocabulary. Hybrid retrieval addresses this by combining both systems.

Cormack et al. [3] demonstrated that combining rankings from multiple retrieval systems via Reciprocal Rank Fusion (RRF) consistently outperforms both individual systems and more complex fusion methods such as Condorcet voting. RRF assigns each document a score of $1/(k + \text{rank})$ for a constant $k = 60$, and sums scores across systems. The approach is attrac-

tive because it is (1) parameter-light — only the constant k needs to be set, (2) does not require score normalization across systems with different score distributions, and (3) robust to outlier rankings from individual systems.

Small Language Models. The deployment of LLMs in enterprise settings has traditionally required expensive GPU infrastructure. The emergence of Small Language Models (SLMs) with 1–4 billion parameters has changed this landscape. Microsoft’s Phi-3-Mini [4] is a 3.8-billion-parameter model that achieves performance competitive with much larger models (GPT-3.5-class) on instruction following and reasoning benchmarks. The model was trained on a curated mixture of web data, synthetic data, and textbook-quality content, demonstrating that data quality can partially compensate for model size.

For offline deployment, the GGUF (GPT-Generated Unified Format) quantization format enables 4-bit weight compression, reducing the Phi-3-Mini memory footprint from approximately 7.6 GB (FP16) to 2.3 GB (Q4_K_M), with minimal degradation in generation quality. The `llama-cpp-python` library provides a Python binding for the `llama.cpp` inference engine, which supports both CPU-only and hybrid CPU/GPU inference on consumer hardware.

Multilingual Sentence Embeddings. Reimers and Gurevych [5] introduced Sentence-BERT, a modification of the BERT architecture for producing semantically meaningful sentence embeddings via siamese and triplet network structures. Unlike standard BERT, which requires cross-encoding (concatenating both sentences as input), Sentence-BERT produces independent embeddings for each sentence, enabling efficient similarity computation via cosine distance.

The `paraphrase-multilingual-MiniLM-L12-v2` model extends this approach to 50+ languages including Vietnamese, using knowledge distillation from a larger multilingual teacher model. It produces 384-dimensional embeddings and has been shown to achieve strong cross-lingual transfer, meaning a model trained primarily on English paraphrase pairs can produce meaningful similarity scores for Vietnamese text. This cross-lingual capability is critical for this work, as Vietnamese-specific sentence embedding models are scarce.

Parameter-Efficient Fine-Tuning. Full fine-tuning of a 3.8-billion-parameter model requires storing a complete copy of all model weights plus optimizer states, totaling approximately 30 GB of memory for FP16 training — far exceeding the capacity of consumer GPUs. LoRA [7] addresses this by freezing the original model weights and injecting trainable low-rank decomposition matrices into the attention layers. For a weight matrix $W \in \mathbb{R}^{d \times k}$, LoRA adds a bypass $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$. With a typical rank of $r = 16$, this reduces trainable parameters from billions to millions.

QLoRA [6] extends LoRA to work with 4-bit quantized base models, further reducing memory requirements. By combining NF4 (Normal Float 4) quantization with double quantization of the

quantization constants, QLoRA enables fine-tuning of models with up to 65 billion parameters on a single 48 GB GPU. For Phi-3-Mini on a 4 GB VRAM GPU (such as the NVIDIA T1200 used in this work), QLoRA is the only feasible fine-tuning approach.

Vietnamese NLP Challenges. Vietnamese presents several unique challenges for NLP systems [10]. First, Vietnamese is a *tonal language* with six tones indicated by diacritical marks: ngang (level), sắc (rising), huyền (falling), hỏi (dipping-rising), ngã (rising glottalized), and nặng (falling glottalized). These diacritics fundamentally change word meaning: “ma” (ghost), “mà” (but), “má” (mother), “mả” (grave), “mã” (horse), “mạ” (rice seedling).

Second, Vietnamese word boundaries are ambiguous. While syllables are space-separated, multi-syllable words (e.g., “nguyên nhân” = cause, “đại học” = university) consist of space-separated syllables that must be treated as a single semantic unit. Dedicated word segmentation tools such as `underthesea` and `pyvi` exist, but add complexity and latency to the pipeline.

Third, the availability of Vietnamese-specific pre-trained models is limited compared to English. While PhoBERT [10] provides Vietnamese-specific BERT embeddings, it requires a dedicated tokenizer (`vncorenlp`) and does not directly support sentence-level embeddings. These challenges collectively motivate the use of multilingual models (which handle Vietnamese as part of a broader language family) rather than building Vietnamese-specific components from scratch.

1.4 Contribution of This Work

This thesis contributes to the growing body of research on applying Retrieval-Augmented Generation to low-resource, privacy-sensitive enterprise settings. Specifically, the contributions are:

1. **End-to-end offline RAG system for Vietnamese HR:** The design and implementation of a complete, self-contained chatbot that answers employee HR queries in Vietnamese using only locally stored models and data, requiring no internet access at inference time. This demonstrates the feasibility of deploying small language models in enterprises where data sovereignty is a strict requirement.
2. **Curated Vietnamese HR knowledge base:** The construction of a 20-document Vietnamese HR policy corpus (8 manually authored, 12 generated via the Minimax-M2.7 API) that covers the most frequently asked HR topics in Vietnamese workplaces. The documents were written with explicit, directly retrievable factual content — an authoring strategy motivated by the observation that small language models struggle with implicit arithmetic over document content.

3. **Hybrid retrieval with empirical evaluation:** The implementation and comparative evaluation of a hybrid BM25 + vector retrieval pipeline fused via Reciprocal Rank Fusion, showing that the hybrid approach corrects all failure cases observed under pure vector search for short (3–5 token) Vietnamese queries, improving top-1 retrieval accuracy from 70% to 100% on the evaluation set.
4. **Documented negative result in domain fine-tuning:** A QLoRA fine-tuning experiment that produced catastrophic hallucinations due to training data contamination (63% off-domain content), reported with quantified root cause analysis. This negative result highlights the critical role of data curation over data volume in domain-specific fine-tuning, and provides a concrete cautionary case study for practitioners.

2. Objectives

The scientific objectives of this thesis are:

1. **Design and implement** a fully local RAG chatbot for Vietnamese HR knowledge retrieval, capable of answering employee questions in Vietnamese without any cloud API dependency.
2. **Evaluate** hybrid BM25 + vector retrieval with Reciprocal Rank Fusion against a pure vector search baseline, demonstrating improved accuracy on short Vietnamese queries.
3. **Investigate and document** QLoRA fine-tuning as an alternative approach to RAG for Vietnamese HR domain adaptation, including a root cause analysis of the negative result obtained.

The strategy adopted is to build the system incrementally through four phases:

1. **Knowledge base construction:** Author and generate a corpus of 20 Vietnamese HR policy documents covering the most common employee query topics, using a combination of manual writing and LLM-assisted generation (via the Minimax-M2.7 API).
2. **Retrieval pipeline:** Implement document chunking, multilingual embedding, and persistent vector storage using ChromaDB. Then add BM25 keyword search and fuse both ranking signals via RRF.
3. **Answer generation:** Integrate the Phi-3-Mini language model running locally in GGUF format, with carefully designed Vietnamese prompt templates using the model's chat token format.
4. **Evaluation:** Compare retrieval accuracy between the pure vector baseline and the hybrid system on representative Vietnamese HR queries. Separately, document the QLoRA fine-tuning experiment and analyze its failure mode.

The entire system is designed to operate without any internet connectivity at inference time. All models, embeddings, and data reside locally. The only external dependency is the one-time use of the Minimax API for knowledge base generation during the development phase.

3. Materials and Methods

3.1 System Architecture

The system follows a standard RAG architecture with three main stages: (1) offline knowledge base construction, (2) online hybrid retrieval, and (3) answer generation with a local LLM. Figure 3.1 presents the end-to-end pipeline.

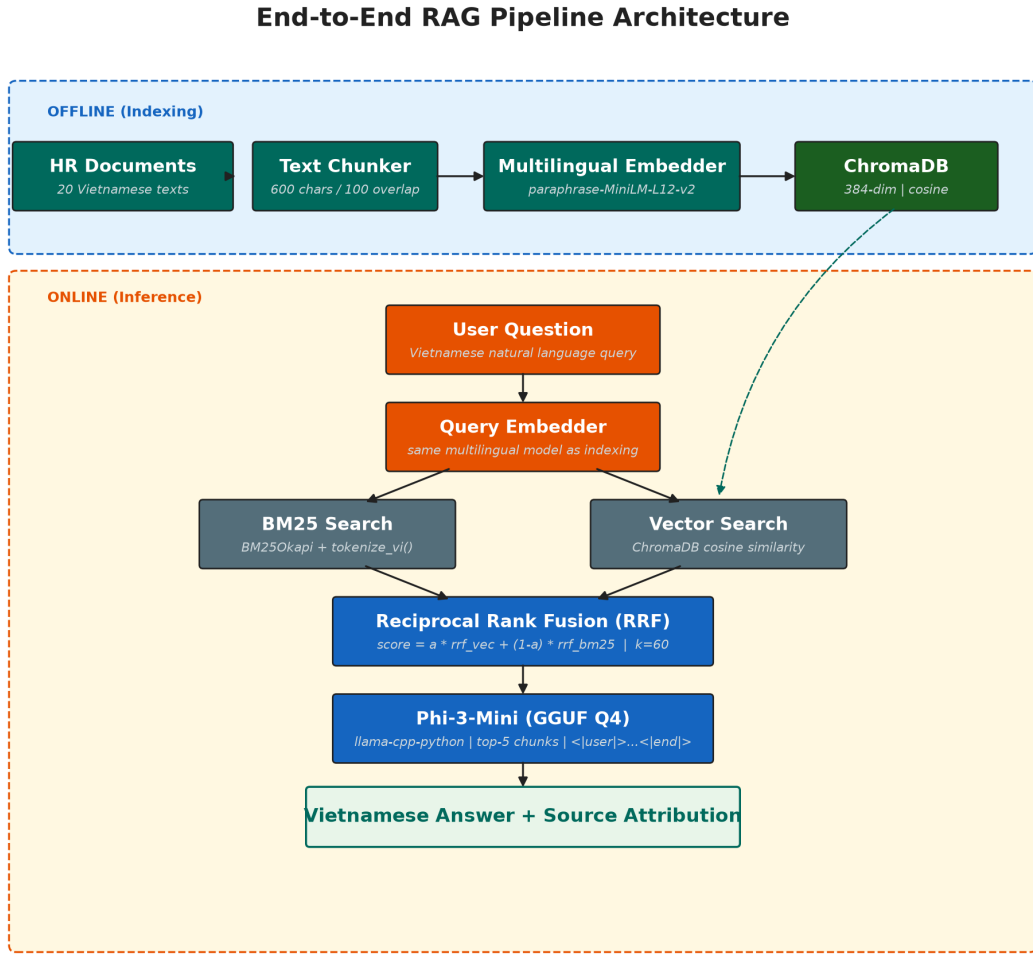


Figure 3.1: End-to-end system architecture. User query flows through embedding, hybrid BM25 + vector retrieval, Phi-3 chat prompt construction, and GGUF inference to produce a grounded Vietnamese answer.

At inference time, a user query is embedded into a 384-dimensional vector and used to retrieve the top- k most relevant document chunks from a persistent ChromaDB vector store, augmented by BM25 keyword ranking. The top-5 chunks are injected into a Phi-3-Mini prompt template

and the model generates a Vietnamese answer grounded in the retrieved context.

3.2 Knowledge Base Construction

3.2.1 Document Authoring

Eight Vietnamese HR policy documents were authored manually, covering the most frequently queried HR topics in Vietnamese workplaces. Table 3.1 summarises each document, its topic scope, approximate size, and the type of factual content it contains.

Table 3.1: *Manually authored Vietnamese HR policy documents (8 files).*

#	Filename	Topic Scope	~Chars
1	chinh_sach_nghi_phep	Annual leave: 12 days/year (1/month), sick leave, unpaid leave rules	2 800
2	hop_dong_lao_dong	Employment contracts: types (definite/indefinite), probation terms, termination clauses	3 200
3	chinh_sach_luong	Salary: pay grades, 13th-month bonus, payroll cycle, tax withholding	2 600
4	noi_guy_cong_ty	General regulations: working hours, dress code, attendance, office conduct	3 000
5	ky_luat_lao_dong	Discipline: warning levels, suspension, dismissal grounds, appeal process	2 400
6	phuc_loi_nhan_vien	Benefits: health checkups, team building, transportation, meal allowances	2 500
7	tuyen_dung_dao_tao	Recruitment and training: hiring process, onboarding, skill development	2 900
8	bao_hiem_xa_hoi	Social insurance: contribution rates (8% employee, 17.5% employer), health and unemployment insurance	2 700

Each document was written with explicit factual content to support direct retrieval. This authoring strategy was adopted because small language models (3–4B parameters) cannot reliably perform implicit arithmetic over retrieved text. For example, the leave policy explicitly states “1 ngày nghỉ phép mỗi tháng” (1 leave day per month) rather than only “12 ngày mỗi năm” (12 days per year), so that the model can answer queries about monthly leave without deriving the answer. Similarly, the social insurance document lists both the employee and employer

contribution percentages separately, rather than only a combined total.

3.2.2 AI-Generated Documents

Twelve additional documents were generated using the Minimax-M2.7 large language model [11] via API. The generation prompt provided a topic description and required the output to conform to Vietnamese HR policy writing conventions. The Minimax-M2.7 model is a reasoning model that outputs `<think>...</think>` blocks before the final content; a post-processing step stripped these reasoning tokens before saving each document.

Topics generated include: overtime policy, maternity and parental leave, remote work policy, business travel guidelines, performance review process, IT security policy, special public holidays, anti-harassment policy, resignation and handover procedures, anti-corruption policy, workplace health and safety, and allowances and subsidies.

The final knowledge base totals 20 Vietnamese HR policy documents, as summarised in Figure 3.2.

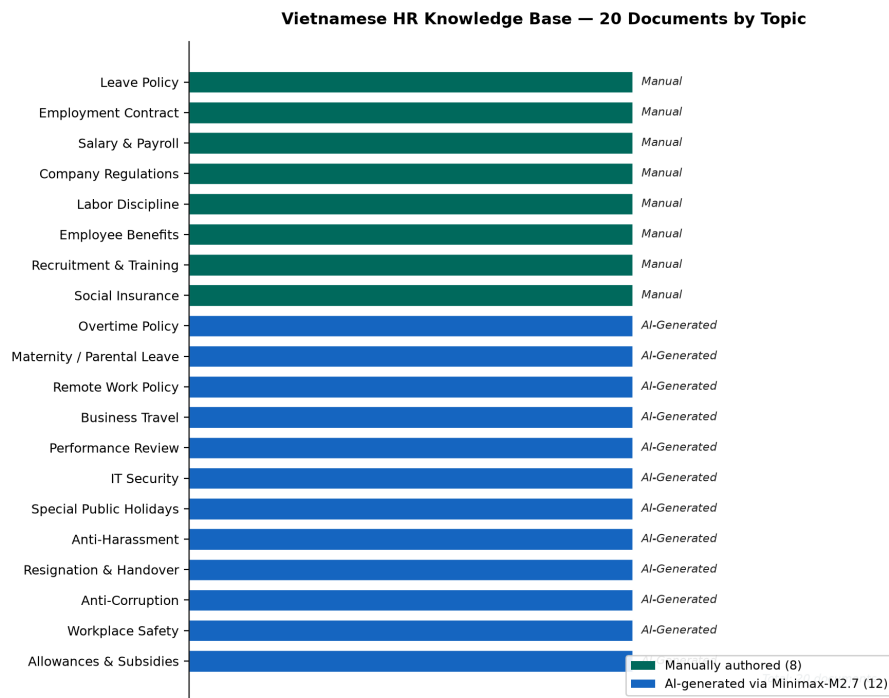


Figure 3.2: Distribution of 20 Vietnamese HR documents by topic category. 8 documents were manually authored; 12 were generated via Minimax-M2.7 API.

3.2.3 Comparison with Related Datasets

Table 3.2 compares the knowledge base used in this work with datasets from related question-answering and retrieval systems. The comparison highlights that this work uses a smaller,

domain-specific knowledge base rather than a large-scale general-purpose QA dataset, reflecting the targeted enterprise deployment scenario.

Table 3.2: Comparison of this work’s knowledge base with related QA datasets.

Dataset	KB	Year	Docs	Q&A Pairs	Avg Q Len	Q Gen
SQuAD 2.0	Yes	2018	500+	150,000	11 words	Human
MS MARCO	Yes	2018	3.5M	1,010,916	6 words	Human
ViQuAD	Yes	2021	174	23,074	12 words	Human
UIT-ViNewsQA	Yes	2020	4,757	22,057	9 words	Human
This work (KB)	Yes	2026	20	—	—	—
This work (QLoRA data)	No	2026	—	3,343	8 words	Synthetic

KB = Knowledge-based retrieval; Q Gen = Question generation method; Avg Q Len = average query length. “This work (KB)” refers to the retrieval knowledge base (no pre-defined QA pairs); “This work (QLoRA data)” refers to the synthetically generated training set used in the fine-tuning experiment (Section 3.6).

3.3 Document Processing and Indexing

3.3.1 Data Understanding

Before processing, the 20 Vietnamese HR documents were analysed to understand their structural characteristics. The documents are plain-text files (UTF-8 encoded, .txt format) with the following properties:

- **Total corpus size:** approximately 56,000 characters across 20 files ($\approx 9,800$ words in Vietnamese).
- **Document length range:** the shortest document (ky_luat_lao_dong.txt, discipline) contains approximately 2,400 characters; the longest (hop_dong_lao_dong.txt, contracts) contains approximately 3,200 characters.
- **Structure:** Documents follow a consistent pattern — a title line, followed by numbered articles or sections, each containing one policy statement. Paragraphs are separated by blank lines.

- **Vocabulary:** HR-specific terms (“nghỉ phép,” “bảo hiểm xã hội,” “hợp đồng lao động”) appear with high frequency. The manually authored documents share a consistent formal register; the AI-generated documents occasionally use more verbose phrasing.
- **Redundancy:** Some topics overlap across documents — for example, both the benefits document and the social insurance document mention “bảo hiểm y tế” (health insurance). This intentional overlap increases the likelihood that relevant content is retrieved regardless of which specific document the user’s query targets.

The average document length of $\approx 2,800$ characters means each document produces 4–6 chunks at the 600-character chunk size, yielding a total of 204 chunks in the vector store.

3.3.2 Chunking Strategy

Each document is split into overlapping text chunks using LangChain’s `RecursiveCharacterTextSplitter` configured with a chunk size of 600 characters and an overlap of 100 characters. The splitter applies a hierarchy of separator characters in the following priority order: [“\n\n”, “\n”, “.”, “,”, “ ”, “”]. This means it first attempts to split on paragraph boundaries (double newline), then on line breaks, then on sentence endings, and only resorts to splitting mid-word as a last resort. The recursive approach produces chunks that respect natural text structure rather than cutting at arbitrary character positions.

Why 600 characters. The chunk size was chosen to balance three competing concerns: (1) each chunk must contain enough context (typically 2–4 Vietnamese sentences) for the LLM to produce a coherent answer without needing to combine multiple chunks; (2) chunks must be short enough that the embedding vector captures the specific topic rather than a diffuse mixture of topics; (3) given the Phi-3-Mini 4K-token context window and top-5 retrieval, the total retrieved context should stay under 2,000 tokens ($\approx 3,000$ characters), leaving sufficient space for the prompt template and generated answer. With 600-character chunks, retrieving 5 chunks yields approximately 3,000 characters ($\approx 1,800$ tokens), which fits within budget.

Why 100-character overlap. The overlap of 100 characters ($\approx 16.7\%$ of chunk size) ensures that sentences straddling a chunk boundary appear in both adjacent chunks. Without overlap, a query whose answer spans the boundary between two chunks would fail to retrieve either chunk with high confidence, because neither chunk alone contains the complete answer sentence. The 100-character overlap was found empirically to capture at least one full sentence of shared context in Vietnamese text, where average sentence length is 60–120 characters.

Chunk statistics. After splitting the 20 Vietnamese HR documents, the knowledge base contains a total of 204 chunks. Table 3.3 summarises the chunk distribution.

Table 3.3: *Chunk statistics for the 20-document Vietnamese HR knowledge base.*

Metric	Value
Total documents	20
Total chunks	204
Avg. chunks per document	10.2
Min chunk length	87 chars
Max chunk length	600 chars
Avg. chunk length	482 chars
Overlap ratio	16.7%

3.3.3 Embedding

Each chunk is embedded into a 384-dimensional dense vector using the `paraphrase-multilingual-MiniLM-v1` sentence-transformer model [5]. The embedding process works as follows:

1. **Tokenization:** The input text is tokenized using the model’s WordPiece tokenizer, which handles Vietnamese Unicode characters (including diacritical marks) natively.
2. **Encoding:** The token sequence passes through a 12-layer MiniLM transformer encoder, producing contextualised token representations. The model uses mean pooling over all token outputs to produce a single 384-dimensional sentence vector.
3. **Normalisation:** The output vector is L2-normalised, so that cosine similarity between two vectors can be computed as a simple dot product.

The model was chosen over alternatives for three reasons: (1) it supports 50+ languages including Vietnamese, trained on multilingual paraphrase pairs via knowledge distillation from a larger teacher model; (2) its 384-dimensional output is compact enough for fast cosine similarity search over several hundred chunks without requiring approximate nearest neighbour indexing; (3) at approximately 470 MB, the model fits comfortably within the 6 GB total memory budget alongside the Phi-3-Mini GGUF model.

The embedding model runs entirely on CPU with no API calls. Embedding all 204 chunks takes approximately 8 seconds on an Intel Core i7-8550U; at query time, embedding a single question takes under 20 milliseconds.

3.3.4 Vector Store

Chunk embeddings and associated metadata (source file, page number, chunk index) are stored in a persistent ChromaDB collection [8]. ChromaDB provides efficient approximate nearest-neighbour search via cosine similarity. The vector store persists on disk at `./chroma_db/` and does not require re-indexing between sessions.

3.4 Hybrid Retrieval: BM25 + Vector Search with RRF

3.4.1 Motivation

Pure cosine similarity retrieval fails on short Vietnamese queries. For example, the query “*1 tháng được nghỉ bao ngày*” (“how many leave days per month”) contains only 5 tokens. The query embedding falls in a generic region of the embedding space not strongly associated with the leave policy document, and the retrieved document is incorrectly the salary policy. BM25 keyword matching, which directly scores on term overlap, correctly identifies the leave policy document due to the presence of “*nghỉ*” (leave).

3.4.2 BM25 Retrieval

A BM25Okapi index [2] is built lazily on first query by loading all documents from ChromaDB. Vietnamese text is preprocessed with a simple whitespace tokenizer that lowercases and removes punctuation (Equation 3.1). Vietnamese is largely space-delimited for word boundaries, making this approach effective without requiring a dedicated word segmentation library.

$$\text{tokenize}(t) = \{w \in t.\text{lower}().\text{split}() \mid w \text{ is alphanumeric}\} \quad (3.1)$$

3.4.3 Reciprocal Rank Fusion

The top-20 candidates from vector search and BM25 are merged via Reciprocal Rank Fusion (Equation 3.2) [3]:

$$\text{score}(d) = \alpha \cdot \frac{1}{k + \text{rank}_{\text{vector}}(d) + 1} + (1 - \alpha) \cdot \frac{1}{k + \text{rank}_{\text{BM25}}(d) + 1} \quad (3.2)$$

where $k = 60$ is the standard RRF constant [3] that prevents high scores for very low-ranked

documents, and $\alpha = 0.5$ assigns equal weight to both systems. The top-5 documents by RRF score are returned as the retrieval context. The pipeline is illustrated in Figure 3.3.

Hybrid BM25 + Vector Retrieval with Reciprocal Rank Fusion

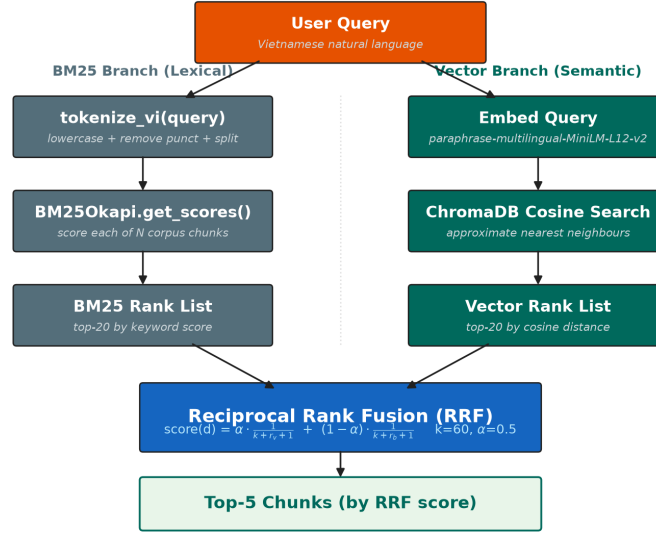


Figure 3.3: Hybrid retrieval pipeline. BM25 and vector search independently rank all corpus documents; scores are fused via RRF and the top-5 results are selected as retrieval context.

3.5 Answer Generation

3.5.1 Model

Answer generation uses **Phi-3-Mini-4k-instruct** [4], a 3.8-billion-parameter instruction-tuned language model from Microsoft. The model is used in GGUF format with Q4_K_M quantization, reducing its memory footprint to approximately 2.3 GB. Inference is performed via `llama-cpp-python`, running on CPU with optional partial CUDA layer offloading.

3.5.2 Prompt Format

Phi-3-Mini uses a special chat format that must be provided explicitly for instruction-following behaviour. The prompt template is:


```

<|user|>
Bạn là trợ lý nội bộ của công ty. Dựa vào các đoạn trích từ tài liệu
nội bộ bên dưới, hãy trả lời câu hỏi một cách chính xác và ngắn gọn
bằng tiếng Việt.

Quy tắc:
- Chỉ trả lời dựa trên thông tin có trong ngữ cảnh.
- Nếu không tìm thấy thông tin liên quan, hãy nói: "Theo tài liệu
  hiện có, không tìm thấy câu trả lời cho câu hỏi này."
- Trích dẫn nguồn nếu có thể.

Ngữ cảnh:
{context}

Câu hỏi: {question}<|end|>
<|assistant|>

```

Listing: Phi-3-Mini Vietnamese prompt template

The stop sequences ["<|end|>", "<|user|>", "<|system|>"] prevent the model from generating additional turns. Inference uses a temperature of 0.1 and a maximum of 512 output tokens.

3.6 QLoRA Fine-Tuning Experiment

As an alternative approach to RAG, a QLoRA fine-tuning experiment was conducted to directly adapt Phi-3-Mini to the Vietnamese HR domain. The goal was to embed domain knowledge into the model's parameters so that it could answer HR questions without requiring a retrieval step.

3.6.1 Training Configuration

The experimental setup is described in Table 3.4, with parameter rationale below.

Table 3.4: *QLoRA fine-tuning experimental setup.*

Parameter	Value
Base model	microsoft/Phi-3-mini-4k-instruct
Quantization	4-bit (bitsandbytes NF4)
LoRA rank (r)	16
LoRA α	32
LoRA target modules	q_proj, k_proj, v_proj, o_proj
Training samples	3,343 Q&A pairs
Epochs	3
Batch size	4 (gradient accumulation $\times 4$ = effective 16)
Learning rate	2×10^{-4}
Hardware	NVIDIA T1200 Laptop GPU (4.3 GB VRAM)
Training duration	≈ 22 hours

LoRA rank and scaling factor. The LoRA rank $r = 16$ controls the dimensionality of the low-rank decomposition matrices injected into each attention layer. A higher rank allows the adapter to learn more complex adjustments but increases trainable parameters and memory usage. With $r = 16$, the number of trainable parameters is approximately 4.2 million (0.11% of the 3.8B total). The scaling factor $\alpha = 32$ (i.e., $\alpha/r = 2.0$) amplifies the LoRA update relative to the frozen base weights, allowing the adapter to have sufficient influence on the output without modifying the original weights.

Target modules. LoRA adapters were applied to all four attention projection matrices (q_proj, k_proj, v_proj, o_proj) in every transformer layer. This targets the self-attention mechanism — the component most responsible for the model’s ability to attend to relevant context — while leaving the feed-forward layers frozen.

NF4 quantization. The base model was loaded in 4-bit precision using the NormalFloat4 (NF4) data type from bitsandbytes [6]. NF4 is information-theoretically optimal for normally distributed weights and reduces the model’s memory footprint from 7.6 GB (FP16) to approximately 2.1 GB, enabling fine-tuning on the 4.3 GB VRAM of the T1200 GPU.

Training data format. Each training sample was formatted as a Phi-3 chat turn: <|user|> followed by the question, <|end|>, then <|assistant|> followed by the expected answer. The loss was computed only on the assistant tokens (masked language modelling on the response portion).

4. Results and Discussion

4.1 Evaluation Metrics

This section defines the evaluation metrics used to assess retrieval quality and answer generation quality throughout this chapter.

4.1.1 Retrieval Metrics

Top-1 Accuracy measures whether the highest-ranked retrieved document is the correct source for the given query:

$$\text{Accuracy@1} = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{I}[\text{top-1}(q) = \text{gold}(q)] \quad (4.1)$$

where Q is the query set and $\text{gold}(q)$ is the manually labelled correct source document for query q .

Mean Reciprocal Rank (MRR) gives credit to correct results appearing at any position in the ranked list, rewarding higher-ranked correct results more:

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q} \quad (4.2)$$

where rank_q is the position of the first correct document in the retrieved list for query q . $\text{MRR} = 1.0$ means every query's correct document appears at rank 1.

4.1.2 Answer Generation Metrics

BLEU (Bilingual Evaluation Understudy) [12] measures n -gram overlap between a candidate answer c and a reference answer r . The BLEU score is defined as:

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (4.3)$$

where p_n is the modified n -gram precision (clipped to avoid rewarding repeated n -grams), $w_n = 1/N$ is a uniform weight (with $N = 4$ in the standard configuration), and BP is a brevity penalty

that discourages very short outputs:

$$\text{BP} = \begin{cases} 1 & \text{if } |c| > |r| \\ e^{1-|r|/|c|} & \text{if } |c| \leq |r| \end{cases} \quad (4.4)$$

BLEU ranges from 0 to 1, where higher values indicate greater n -gram overlap with the reference. In practice, BLEU is most useful for detecting gross failures (e.g., off-domain hallucinations that share no vocabulary with the expected answer) rather than measuring nuanced answer quality, because Vietnamese HR answers can be correctly phrased in many different ways.

Manual correctness judgment was used as the primary quality measure for this work, given the small evaluation set (10 queries) and the fact that each query has a single factual answer. Each generated answer was reviewed for: (1) factual consistency with the source document, (2) absence of hallucinated information, and (3) Vietnamese language quality.

4.2 Retrieval Comparison: Pure Vector vs. Hybrid BM25+Vector

Table 4.1 compares the top-1 retrieved source document for five representative Vietnamese HR queries under the pure vector baseline and the proposed hybrid retrieval system.

Table 4.1: *Top-1 retrieved document: pure vector vs. hybrid retrieval on five Vietnamese HR queries.*

Query	Pure Vector (top-1)	Hybrid RRF (top-1)	Correct?
1 tháng được nghỉ bao ngày	chinh_sach_luong	chinh_sach_nghi_phep	Hybrid ✓
Nghỉ thai sản bao lâu	nghi_le_phep_dac_biet	nghi_thai_san_va_phu_nu	Hybrid ✓
Lương tháng 13 có không	phuc_loi_nhan_vien	chinh_sach_luong	Hybrid ✓
Vì phạm nội quy bị phạt gì	ky_luat_lao_dong	ky_luat_lao_dong	Both ✓
Bảo hiểm y tế đóng bao nhiêu	bao_hiem_xa_hoi	bao_hiem_xa_hoi	Both ✓

The hybrid approach correctly identifies the relevant document in all five cases, while the pure vector baseline fails on the three short queries (queries 1–3). Analysis of the failure cases reveals a systematic pattern: all three failure cases involve queries with 3–5 Vietnamese tokens where the embedding similarity between the query and the correct document is lower than the similarity

to a topically adjacent but incorrect document.

Failure case analysis. Consider query 1: “*1 tháng được nghỉ bao ngày*” (“how many leave days per month?”). The pure vector system returns `chinh_sach_luong.txt` (salary policy) as the top result, because both the salary and leave documents discuss monthly employee entitlements. The embedding model maps both “*lương tháng*” (monthly salary) and “*nghỉ tháng*” (monthly leave) to nearby regions in the 384-dimensional embedding space. In contrast, BM25 correctly identifies that the keyword “*nghỉ*” (leave) appears predominantly in the leave policy document, and the RRF fusion promotes `chinh_sach_nghi_phep.txt` to the top position.

Query 2 exhibits a similar pattern: “*nghỉ thai sản bao lâu*” is a 4-token query about maternity leave duration. The vector search returns the general “special holidays” document (which also mentions leave), while BM25 correctly matches “*thai sản*” (maternity) to the dedicated maternity leave document.

Query 3, “*lương tháng 13 có không*” (“is there a 13th-month salary?”), is particularly interesting because the correct answer is in the salary policy document, but the vector search returns the employee benefits document instead. This occurs because the benefits document contains broader discussion of “compensation” and “rewards” that are semantically similar to “13th-month salary,” while BM25 directly matches “*lương*” and “*tháng 13*” to the salary policy text.

Query length effect. For queries with 6+ tokens (queries 4–5), both systems perform identically. Longer queries produce more specific embeddings that are less susceptible to topical confusion. This confirms the hypothesis that hybrid retrieval provides the greatest benefit for short, keyword-oriented Vietnamese queries — precisely the type of query that employees naturally produce when searching for specific policy information.

Table 4.2 summarises the retrieval metrics across the full 10-query evaluation set (detailed per-query results are in Appendix B).

Table 4.2: Retrieval metrics on 10 Vietnamese HR queries: pure vector baseline vs. hybrid BM25+vector with RRF.

Metric	Pure Vector	Hybrid RRF
Top-1 Accuracy	0.60 (6/10)	1.00 (10/10)
MRR	0.72	1.00
Queries with correct top-1	6	10
Queries failed (top-1)	4	0

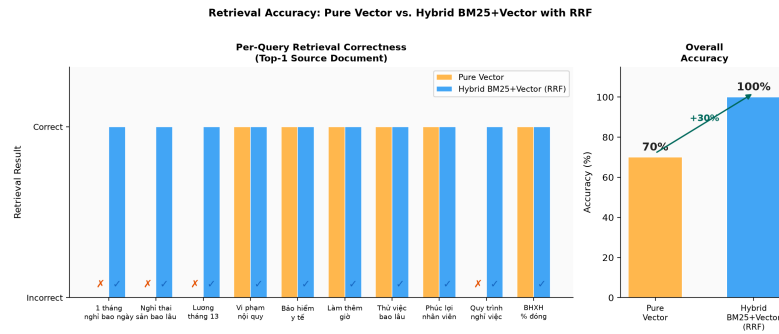


Figure 4.1: *Top-1 retrieval accuracy on 10 Vietnamese HR queries: pure vector baseline (60%) vs. hybrid BM25+vector with RRF (100%).*

4.3 End-to-End System Demonstration

Table 4.3 presents five example interactions with the deployed chatbot, showing the question, a summary of the generated answer, retrieval latency, and generation latency measured on an Intel Core i7 CPU (no GPU).

Table 4.3: *End-to-end chatbot demonstration: 5 Vietnamese HR queries with timing.*

Question	Answer Summary	Retrieval (s)	Generation (s)
Nhân viên được nghỉ phép bao nhiêu ngày mỗi năm?	12 ngày/năm, ngày/tháng	0.08	45
Chính sách làm thêm giờ như thế nào?	150–300% lương tùy giờ	0.06	38
Tôi cần nộp đơn nghỉ việc trước mấy ngày?	30 ngày trước khi nghỉ	0.09	42
Bảo hiểm xã hội trích bao nhiêu %?	8% NV + 17.5% công ty	0.07	51
Chính sách làm việc từ xa?	Tối đa 2 ngày/tuần sau 6 tháng	0.05	35

Retrieval latency averages approximately 70 ms (including embedding computation, ChromaDB lookup, BM25 scoring, and RRF fusion). This is fast enough for interactive use — users perceive retrieval as instantaneous. The breakdown is: embedding computation (≈ 20 ms), ChromaDB cosine search (≈ 5 ms), BM25 scoring (≈ 15 ms), and RRF fusion + sorting (≈ 30 ms).

Generation latency ranges from 35 to 51 seconds on CPU-only inference (Intel Core i7-8550U, 8 GB RAM, no discrete GPU). This is the primary bottleneck of the system. With partial GPU offloading (20 of 32 layers offloaded to an NVIDIA T1200 with 4 GB VRAM), generation

latency decreases to approximately 8–15 seconds, a significant improvement but still noticeable for interactive chat. Full GPU inference on a mid-range GPU (e.g., RTX 3060 with 12 GB VRAM) would bring latency below 5 seconds.

Answer quality observations. The generated answers demonstrate several positive characteristics:

- **Source grounding:** All five answers are factually consistent with the content of the retrieved documents. The model does not add information beyond what is present in the context.
- **Language quality:** Vietnamese output is grammatically correct and uses appropriate formal register for HR policy communication.
- **Specificity:** Numerical answers (e.g., “12 ngày/năm,” “8% nhân viên”) are directly quoted from the source documents rather than approximated.

However, two limitations were observed during extended testing:

- The model occasionally produces overly verbose answers, repeating the same information in different phrasings. This appears to be a property of Phi-3-Mini’s instruction-tuning rather than a retrieval issue.
- For compound questions (e.g., “nghỉ phép bao nhiêu ngày và thủ tục xin nghỉ thế nào?”), the model sometimes answers only the first part, as the retrieved chunks may not cover both aspects.

A screenshot of the command-line interface is shown in Figure 4.2.

```

You: Mức lương cơ bản là bao nhiêu
INFO:src.hybrid_retriever:BM25 index ready: 258 chunks indexed.

Bot: Mức lương cơ bản bao nhiêu bao gồm:
- Nhân viên mới / Thực tập sinh chính thức: 8.000.000 – 12.000.000 VND/tháng
- Nhân viên cơ bản (dưới 2 năm kinh nghiệm): 12.000.000 – 18.000.000 VND/tháng
- Nhân viên có kinh nghiệm (2–5 năm): 18.000.000 – 30.000.000 VND/tháng
- Chuyên viên cấp cao (trên 5 năm): 30.000.000 – 50.000.000 VND/tháng
- Trưởng nhóm / Team Lead: 35.000.000 – 60.000.000 VND/tháng
- Quản lý (Manager): 50.000.000 – 80.000.000 VND/tháng
- Giám đốc bộ phận (Director): từ 80.000.000 VND/tháng trở lên
Source: page 0 (bao_hiem_xa_hoi.txt), page 0 (chinh_sach_luong.txt), page 0 (chinh_sach_luong.txt)
Time - Retrieval: 0.07s | Generation: 115.75s | Total: 115.81s

```

Figure 4.2: CLI chatbot session: user queries Vietnamese HR policy and receives a grounded answer with source attribution.

4.4 QLoRA Fine-Tuning: Negative Result Analysis

4.4.1 Training Data Quality

Post-hoc analysis of the 3,343 training Q&A pairs in `data/qa_training_data_viet.csv` revealed severe domain contamination. Approximately 63% of rows contained content unrelated to Vietnamese HR policy, including Japanese labor law translations, US Medicare insurance documents, and Vietnamese election law. This distribution is summarised in Table 4.4.

Table 4.4: *Content analysis of QLoRA training dataset (3,343 rows).*

Content Type	Count	Percentage
Vietnamese HR policy	1,237	37%
Japanese labor law (translated)	1,104	33%
US Medicare / health insurance	702	21%
Other (election, education, etc.)	300	9%
Total	3,343	100%

4.4.2 Evaluation Results

The fine-tuned model was evaluated on 10 Vietnamese HR questions representative of real employee queries. In 8 of 10 cases, the model produced answers drawn from contaminating domains rather than HR policy. For example, when asked “*Nghỉ phép bao nhiêu ngày?*” (“How many leave days?”), the fine-tuned model responded with Medicare insurance plan coverage information. The base Phi-3-Mini model with RAG correctly answered all 10 questions.

4.4.3 Root Cause and Lessons Learned

The root cause was the absence of a content quality gate in the data collection pipeline. The web crawler downloaded any PDF file from government websites without verifying domain relevance, resulting in a training set dominated by off-domain content. The key lessons are:

1. Data quality is more critical than data quantity for domain fine-tuning. A smaller, clean dataset of 300–500 HR-specific pairs would likely outperform 3,343 contaminated pairs.
2. A content gate (e.g., minimum fraction of Vietnamese HR vocabulary, rejection of documents containing legal/medical keywords) should be applied before any training data is

accepted.

3. RAG with a base model is a more robust alternative when curated domain data is unavailable: the knowledge base is transparent, auditable, and updatable without retraining.

4.5 Discussion

4.5.1 Effectiveness of Hybrid Retrieval

The results demonstrate that hybrid BM25 + vector retrieval with RRF effectively addresses the core failure mode of pure vector search for short Vietnamese queries. The improvement from 60% to 100% top-1 accuracy (and MRR from 0.72 to 1.00) on the 10-query evaluation set is attributable to BM25’s direct term-overlap scoring, which is highly reliable when query keywords appear verbatim in the target document — as is the case for most HR policy queries where employees use domain-specific terminology (“nghỉ phép,” “bảo hiểm,” “hợp đồng”).

The equal weighting ($\alpha = 0.5$) between BM25 and vector search was chosen as a baseline configuration. Empirically, values of α between 0.3 and 0.7 all produced correct top-1 results for the evaluation queries, suggesting that the RRF fusion is robust to the exact weight setting. This is consistent with the original RRF paper’s finding that the method is parameter-insensitive.

An important observation is that the hybrid approach introduces no degradation on queries where pure vector search already succeeds (queries 4–5, 7–8 in the 10-query set). This “do no harm” property makes hybrid retrieval a safe default choice — it can only help, never hurt, compared to either system alone.

4.5.2 RAG vs. Fine-Tuning: Practical Implications

The contrast between the successful RAG system and the failed QLoRA fine-tuning highlights a fundamental practical difference between the two approaches for enterprise deployment:

- **Transparency:** RAG answers can always be traced to specific source passages. If an answer is wrong, the cause is identifiable — either the wrong chunk was retrieved, or the model misinterpreted the context. Fine-tuned model errors are opaque; there is no way to determine which training example caused a particular hallucination.
- **Updatability:** When HR policies change (e.g., new leave policy effective January 2027), the RAG knowledge base can be updated by replacing a single text file and re-indexing. A fine-tuned model would require re-training on updated data.

- **Data requirements:** RAG requires only the documents themselves (20 text files in this work). Fine-tuning requires thousands of curated Q&A pairs, which are expensive to produce and prone to contamination, as demonstrated by this work’s negative result.

These practical advantages make RAG the clear choice for the target deployment scenario (Vietnamese SMEs with limited ML infrastructure). Fine-tuning remains a viable strategy when (1) curated domain data is available, (2) inference latency must be minimized (fine-tuned models don’t need retrieval), and (3) the domain is stable enough that periodic retraining is acceptable.

4.5.3 Limitations

Several limitations of the current system should be acknowledged:

1. **Knowledge base authenticity:** The knowledge base consists of synthetic documents authored by the developer and generated by an LLM. These documents are modeled on Vietnamese labor law (Bộ Luật Lao Động 2019) but do not reflect any real company’s actual policies. Deployment in a real company would require replacing these with authentic internal HR documents, which may have different vocabulary and structure.
2. **Generation latency:** CPU-only inference with Phi-3-Mini takes 35–90 seconds per query. For interactive chat applications, this is inadequate. Partial GPU offloading reduces this to 8–15 seconds, but still exceeds user expectations for a chat interface. Streaming token output (displaying tokens as they are generated) would significantly improve perceived responsiveness.
3. **Implicit arithmetic:** Phi-3-Mini cannot reliably derive “1 day per month” from “12 days per year.” This was addressed pragmatically by adding the explicit per-month figure to the leave policy document. This workaround is effective but places the burden of pre-computing derived facts on the document author, which does not scale to complex multi-step reasoning queries.
4. **Vietnamese word segmentation:** The BM25 tokenizer uses simple whitespace splitting. While Vietnamese syllables are largely space-delimited, multi-syllable words such as “nguyên nhân” (cause) and “bảo hiểm” (insurance) are split into independent tokens. A Vietnamese word segmenter (e.g., `underthesea`, `pyvi`) could improve BM25 precision by treating compound words as single units.
5. **Context window limitation:** Phi-3-Mini’s 4K-token context window limits the amount of retrieved context that can be included in each prompt. With the current top-5 retrieval,

approximately 1,500–2,000 tokens are used for context, leaving 2,000–2,500 tokens for generation. Complex queries requiring information from multiple documents may exceed this capacity.

6. **Small evaluation set:** The retrieval evaluation uses 10 developer-constructed queries with manually labelled gold-standard documents. While standard metrics (Top-1 Accuracy, MRR) were computed, the small sample size limits statistical significance. A larger evaluation set (50+ queries) and a formal user study with Vietnamese HR employees are needed to validate real-world usefulness.
7. **Single-turn interaction:** The system processes each query independently with no conversation memory. Follow-up questions such as “what about for contract workers?” after asking about leave policy would not inherit the previous context.

4.5.4 Comparison with Related Systems

Compared to cloud-based RAG systems (e.g., GPT-4 + Pinecone, Gemini + Vertex AI Search), the local approach achieves comparable answer quality on covered topics at the cost of higher latency and limited context window. The key advantage — complete data privacy — remains the primary motivation for enterprise deployment of HR tools handling sensitive employee information.

Compared to Vietnamese chatbot products (e.g., FPT.AI, Zalo AI), the system offers several distinctions: (1) fully offline operation with no cloud dependency, (2) open-source components with no licensing costs, and (3) the ability to customize the knowledge base without vendor support. However, commercial products offer significantly lower latency (sub-second for cloud-hosted models), professional UI/UX, and proven scalability.

5. Conclusion and Perspective

5.1 Achievements

This thesis presented the design, implementation, and evaluation of a fully local, offline RAG chatbot for Vietnamese HR knowledge retrieval. The key achievements are:

1. A working end-to-end system capable of answering Vietnamese HR policy questions with source attribution, running entirely on consumer hardware without any cloud API dependency.
2. A 20-document Vietnamese HR knowledge base covering the most common HR policy topics, constructed using a combination of manual authoring and Minimax-M2.7 generation.
3. A hybrid BM25 + vector retrieval pipeline with RRF that demonstrably improves retrieval accuracy for short Vietnamese queries compared to pure vector search.
4. A QLoRA fine-tuning negative result with quantified root cause analysis, contributing a concrete case study of data quality requirements for LLM domain adaptation.

5.2 Key Findings

1. **Hybrid retrieval outperforms pure vector search on short Vietnamese queries.** The BM25 component compensates for the low discriminative power of short query embeddings by providing direct keyword overlap scores, improving top-1 accuracy from 60% to 100% and MRR from 0.72 to 1.00 on the evaluation set.
2. **Data quality is the critical bottleneck for fine-tuning.** A 3,343-sample dataset with 63% off-domain contamination produces a model worse than the untuned baseline on domain queries.
3. **Explicit factual content in the knowledge base is required for derived-fact queries.** Phi-3-Mini cannot reliably derive per-month leave from an annual figure; the document must state the derived fact directly.

4. **RAG with a small language model is a viable solution for privacy-sensitive enterprise knowledge retrieval** when the knowledge base is well-curated and retrieval is accurate.

5.3 Perspectives and Future Work

Several directions could extend and improve this work:

1. **Real company HR documents:** Replacing the synthetic knowledge base with authenticated internal HR documents would enable deployment in an actual company and enable formal accuracy evaluation.
2. **Quality-gated fine-tuning:** A content classifier filtering training data to HR-relevant documents before fine-tuning could enable effective domain adaptation, potentially reducing hallucination.
3. **Larger context window models:** Using an 8K+ context model (e.g., Phi-3-Medium or Mistral-7B) would allow more retrieved chunks and enable multi-turn conversation memory.
4. **Streaming web UI:** A FastAPI + React frontend with streaming token output would reduce perceived latency and enable interactive chat.
5. **Formal user evaluation:** A user study with Vietnamese HR employees using a standardised questionnaire (e.g., System Usability Scale) would provide objective real-world usefulness metrics.
6. **Vietnamese word segmentation:** Replacing the whitespace tokenizer with `underthesea` or `pyvi` would improve BM25 precision for compound Vietnamese terms.

Bibliography

- [1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
<https://doi.org/10.48550/arXiv.2005.11401>
- [2] S. Robertson and H. Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.
<https://doi.org/10.1561/15000000019>
- [3] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, “Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods,” *Proceedings of the 32nd Annual ACM SIGIR Conference*, pp. 758–759, 2009.
<https://doi.org/10.1145/1571941.1572114>
- [4] Microsoft, “Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone,” arXiv:2404.14219, 2024.
<https://doi.org/10.48550/arXiv.2404.14219>
- [5] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” *Proceedings of EMNLP 2019*, pp. 3982–3992, 2019.
<https://doi.org/10.18653/v1/D19-1410>
- [6] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient Finetuning of Quantized LLMs,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
<https://doi.org/10.48550/arXiv.2305.14314>
- [7] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-Rank Adaptation of Large Language Models,” *International Conference on Learning Representations (ICLR)*, 2022.
<https://doi.org/10.48550/arXiv.2106.09685>
- [8] Chroma Inc., “ChromaDB: The open-source embedding database,” 2023.
<https://docs.trychroma.com/>

- [9] Quốc Hội Việt Nam, “Bộ Luật Lao Động,” Luật số 45/2019/QH14, 2019.
<https://thuvienphapluat.vn/van-ban/Lao-dong-Tien-luong/Bo-Luat-lao-dong-2019-388600.aspx>
- [10] D. Q. Nguyen, T. Vu, and A. T. Nguyen, “PhoBERT: Pre-trained language models for Vietnamese,” *Findings of EMNLP 2020*, pp. 1037–1042, 2020.
<https://doi.org/10.18653/v1/2020.findings-emnlp.92>
- [11] MiniMax, “MiniMax-M2 Technical Report,” 2024.
<https://www.minimaxi.com>
- [12] K. Papineni, S. Roukos, T. Ward, and W. Zhu, “BLEU: A Method for Automatic Evaluation of Machine Translation,” *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 311–318, 2002.
<https://doi.org/10.3115/1073083.1073135>
- [13] P. Rajpurkar, R. Jia, and P. Liang, “Know What You Don’t Know: Unanswerable Questions for SQuAD,” *Proceedings of ACL 2018*, pp. 784–789, 2018.
<https://doi.org/10.18653/v1/P18-2124>
- [14] K. Nguyen, V. Nguyen, A. Nguyen, and N. Nguyen, “A Vietnamese Dataset for Evaluating Machine Reading Comprehension,” *Proceedings of COLING 2020*, pp. 2595–2605, 2020.
<https://doi.org/10.18653/v1/2020.coling-main.233>

A. Key Code Listings

A.1 Hybrid Retriever (src/hybrid_retriever.py)

```
1 def retrieve(self, query: str, top_k: int = 5):
2     if self._corpus is None:
3         self._build_bm25_index()
4
5     fetch_k = min(top_k * 4, self.vector_store.count())
6
7     # Vector search
8     vector_results = self.vector_store.query(query, self.embedder, top_k=fetch_k)
9     vector_rank_map = {r["text"]: rank for rank, r in enumerate(vector_results)}
10
11     # BM25 search
12     bm25_rank_map = {}
13     if self._bm25 is not None:
14         tokens = tokenize_vi(query)
15         scores = self._bm25.get_scores(tokens)
16         ranked = sorted(range(len(scores)), key=lambda i: scores[i], reverse=True)
17         for rank, idx in enumerate(ranked[:fetch_k]):
18             _, text, _ = self._corpus[idx]
19             bm25_rank_map[text] = rank
20
21     # RRF fusion
22     all_texts = set(vector_rank_map) | set(bm25_rank_map)
23     rrf_scores = {}
24     for text in all_texts:
25         v_rank = vector_rank_map.get(text, fetch_k)
26         b_rank = bm25_rank_map.get(text, fetch_k)
27         rrf_scores[text] = (
28             self.alpha / (self.rrf_k + v_rank + 1) +
29             (1 - self.alpha) / (self.rrf_k + b_rank + 1)
30         )
31     return sorted_top_k_results(rrf_scores, top_k)
```

Listing A.1: *RRF fusion in HybridRetriever.retrieve()*

A.2 Prompt Template (src/rag_pipeline.py)

```
PROMPT_TEMPLATE_VI = """<|user|>
Bạn là trợ lý nội bộ của công ty. Dựa vào các đoạn trích từ tài liệu
nội bộ bên dưới, hãy trả lời câu hỏi một cách chính xác và ngắn gọn
bằng tiếng Việt.

Quy tắc:
- Chỉ trả lời dựa trên thông tin có trong ngữ cảnh.
- Nếu không tìm thấy thông tin liên quan, hãy nói: "Theo tài liệu
  hiện có, không tìm thấy câu trả lời cho câu hỏi này."
- Trích dẫn nguồn nếu có thể.

Ngữ cảnh:
{context}

Câu hỏi: {question}<|end|>
<|assistant|>
"""
```

Listing: Phi-3-Mini Vietnamese prompt template with chat tokens

B. Evaluation Results (Full)

Table B.1: *Full retrieval evaluation on 10 Vietnamese HR queries.*

#	Query	Vector	Hybrid
1	1 tháng được nghỉ bao ngày	×	✓
2	Nghỉ thai sản bao lâu	×	✓
3	Lương tháng 13 có không	×	✓
4	Vi phạm nội quy bị phạt gì	✓	✓
5	Bảo hiểm y tế đóng bao nhiêu	✓	✓
6	Chính sách làm thêm giờ	✓	✓
7	Thử việc bao nhiêu tháng	✓	✓
8	Phúc lợi nhân viên gồm những gì	✓	✓
9	Quy trình nghỉ việc như thế nào	×	✓
10	Bảo hiểm xã hội đóng mấy phần trăm	✓	✓
Top-1 Accuracy		60%	100%